DO NOW — RECALL

# Karnaugh maps and simplification

Name: _____ Class: _____ Date: _____

**1. Simplify ¬(A ∨ B).**

**2. Keys 7 and 12 hash to index 2 in a table. What is that called?**

**3. What is 1 XOR 1?**

**4. If A changes from 0 to 1 while B is always 1, which variable is constant?**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

¬A ∧ ¬B.

### 2. Model answer

A collision.

### 3. Model answer

0.

### 4. Model answer

B, at value 1.

### Vocabulary — know these terms

Karnaugh map — Grid for simplifying Boolean functions using adjacent combinations.

Gray code — Ordering in which adjacent binary labels differ by one bit.

group — A rectangle containing 1, 2, 4, 8, … adjacent 1-cells.

constant variable — A variable with the same value in all cells of a group.

wraparound — The opposite edges of a Karnaugh map are adjacent.

NEW LEARNING

# Read, annotate and explain

### Read the labels

Rows represent A, and columns use Gray order BC: 00, 01, 11, 10.

### Group neighbouring ones

The four 1s form one 2×2 rectangle. Adjacency across rows is allowed.

### Keep what is constant

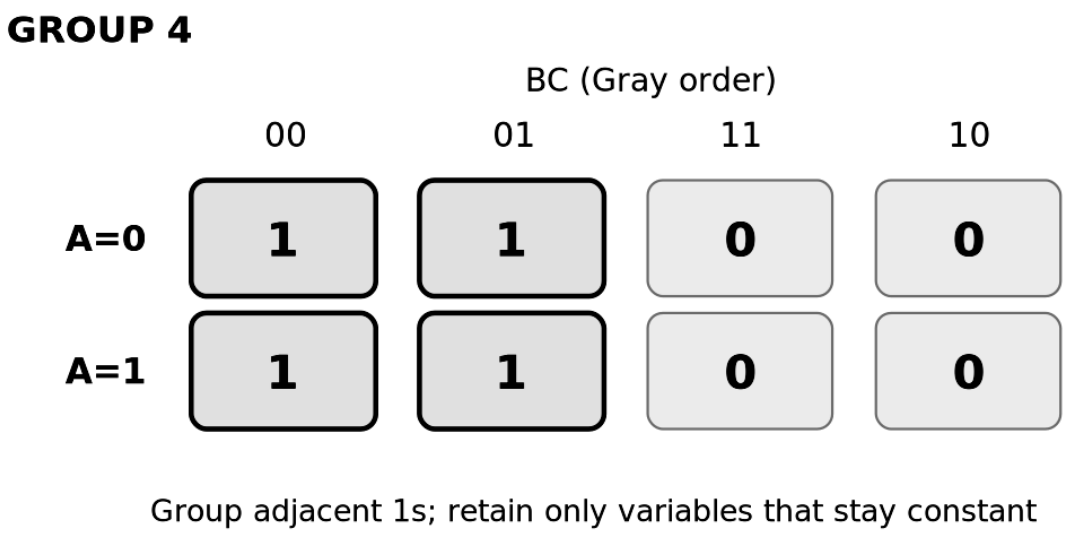
A and C change through the group, but B remains 0. So the simplified result is ¬B.

### Explain one step in your own words, using the correct technical terms.

# New Learning · original lesson diagrams

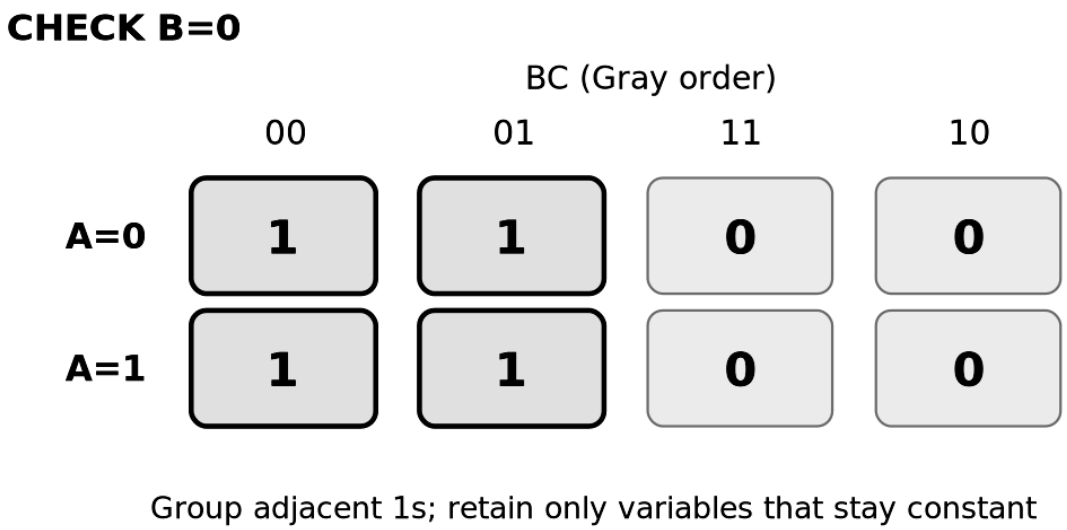
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




GROUP 4 Karnaugh map, columns 00, 01, 11, 10, rows A=0 and A=1

**New Learning**




CHECK B=0 Karnaugh map, columns 00, 01, 11, 10, rows A=0 and A=1

I DO — WORKED EXAMPLE

# Follow the worked method

**Four 1s fill the entire A=0 row. Simplify the output.**

### Predict and explain

B and C each vary across the row; neither is needed in the expression.

### Check and extend

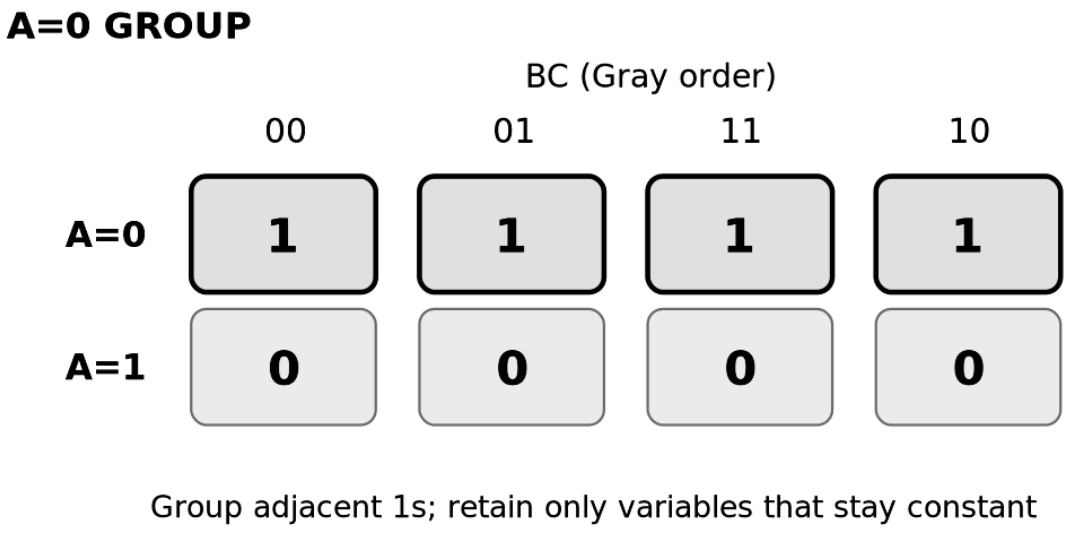
A stays 0 in the whole group. The simplified term is ¬A.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Four 1s fill the entire A=0 row. Simplify the output.**

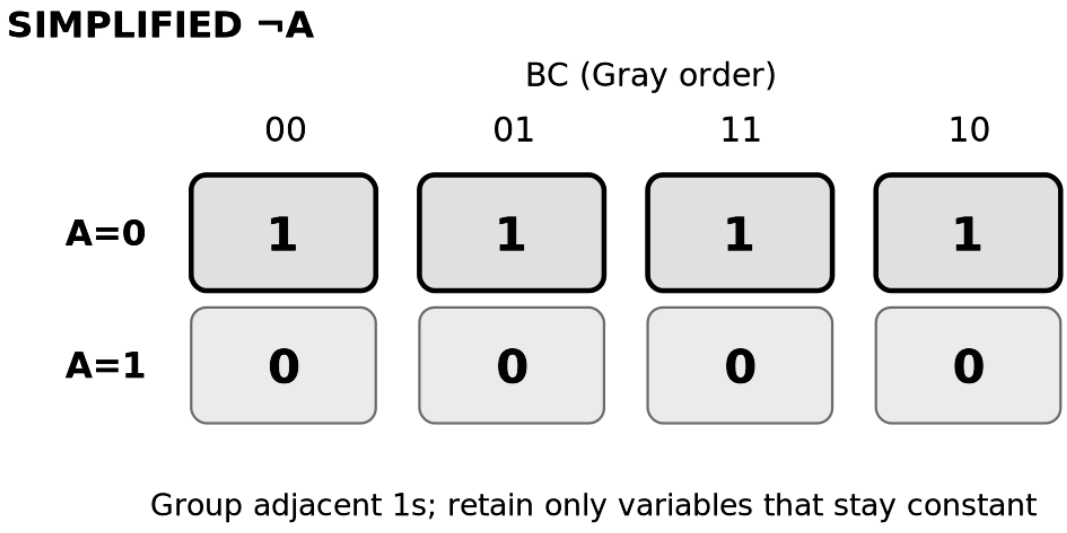
B and C each vary across the row; neither is needed in the expression.




A=0 GROUP Karnaugh map, columns 00, 01, 11, 10, rows A=0 and A=1

**Four 1s fill the entire A=0 row. Simplify the output.**

B and C each vary across the row; neither is needed in the expression.




SIMPLIFIED ¬A Karnaugh map, columns 00, 01, 11, 10, rows A=0 and A=1

**WE DO — GUIDED PRACTICE**

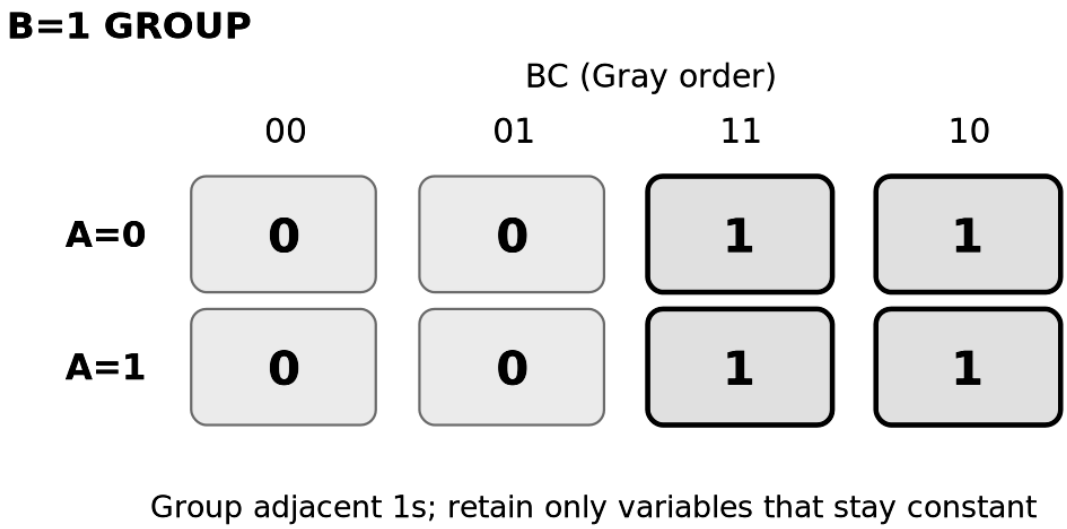
# Try → Discuss → Improve

# We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Four 1s fill columns BC=11 and BC=10 across both rows. Simplify.**

A changes between rows, and C changes between columns 11 and 10.




B=1 GROUP Karnaugh map, columns 00, 01, 11, 10, rows A=0 and A=1

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

### Guided 1: Predict and explain A changes between rows, and C changes between columns 11 and 10.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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**YOU DO — INDEPENDENT PRACTICE**

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Why do we group neighbouring 1s in a Karnaugh map?

### Question 2

The whole A=1 row contains four 1s. Which simplified term remains?

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Columns BC=00 and BC=10 are joined by wraparound adjacency. Four 1s occupy both these columns. Simplify.

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**A group of four 1s can remove two changing variables.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

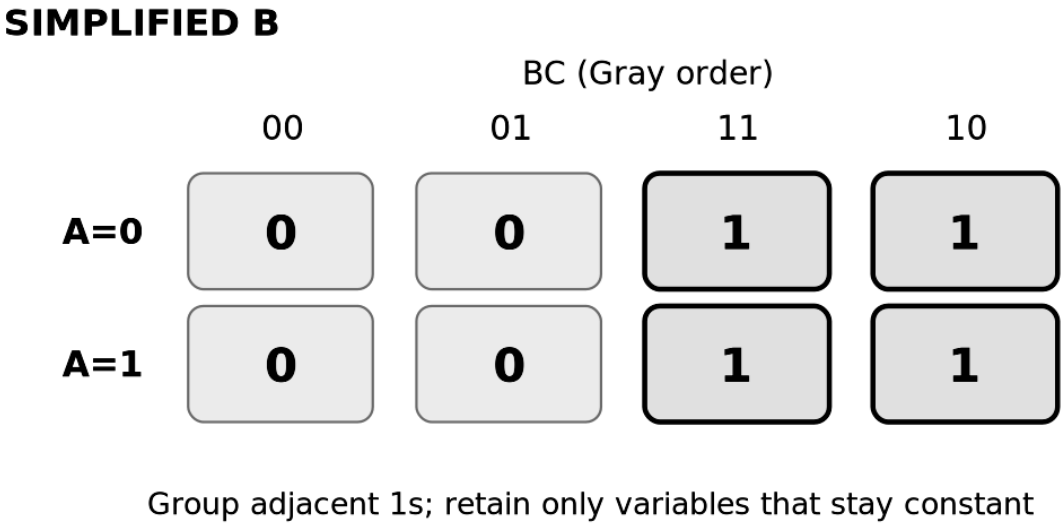
Next step: revisit the model or a relevant revision-guide section if needed.

# We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Four 1s fill columns BC=11 and BC=10 across both rows. Simplify.**

A changes between rows, and C changes between columns 11 and 10.




SIMPLIFIED B Karnaugh map, columns 00, 01, 11, 10, rows A=0 and A=1

# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Check and extend

B remains 1, so the simplified term is B.

IMPROVE: Explain one correction to your own reasoning.